Immigration attitudes: Economic and cultural threat

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In general, do people (in developed countries) support or oppose increases in immigration? On the one hand, studies have shown that immigrants contribute to economic productivity and foster positive cultural exchanges. On the other hand, anecdotal evidence suggests that the public is resistant to large influxes of immigrants and prefers the status quo.

So why might citizens be anti-immigrant? According to economic theory, citizens might worry about additional competition in the labor market and thus oppose increases in immigrant numbers. This argument is known as the *economic threat* hypothesis. Yet, most empirical tests of the economic threat hypothesis have failed to support this argument, and, in fact, people often support policies that may be against their own economic interests.

An alternative hypothesis is that immigration attitudes are influenced by various deep-rooted ethnic and cultural stereotypes. This argument is known as the *cultural threat* hypothesis. Most research suggests that cultural explanations are better predictors of anti-immigration attitudes than economic ones. Given the prominence of workers' economic concerns in the political discourse, coupled with cultural considerations, how can we reconcile these approaches?

This exercise is based in part on Malhotra, N., Margalit, Y. and Mo, C.H., 2013. "Economic Explanations for Opposition to Immigration: Distinguishing between Prevalence and Conditional Impact." *American Journal of Political Science*, Vol. 38, No. 3, pp. 393-433.

In this article, the authors argue that job competition is *conditional* (i.e., not always prevalent) and its impact may dependent on the context (i.e., in selected industries, job competition may be quite sizable; in others, less so). In other words, if we estimate the direct effect of economic threat on immigration attitudes, we'll likely find a small and non-significant result. Instead, economic threat may be *moderated* by other factors such as the gender of the immigrants in question. This nuanced approach requires us to model the data using an interaction term (i.e., the product of two variables; e.g., X1 times X2).

To test their expectations, they conduct a survey of Americans' attitudes toward high-skilled H-1B visas. Most H-1B visas in the US are currently occupied by Indian immigrants, who are skilled but ethnically distinct from the majority of Americans, which enables the authors to measure a specific skill set (high technology) that is threatened by a particular type of immigrant (H-1B visa holders). The dataset `immig.csv` has the following variables:

Name	Description
`age`	Age (in years)
`female`	`1` indicates female; `0` indicates male
`employed`	`1` indicates employed; `0` indicates unemployed
`nontech.whitcol`	`1` indicates non-tech, white-collar worker; `0` otherwise (e.g., lawyer)
`tech.whitcol`	`1` indicates high-technology worker, `0` otherwise
`expl.prejud`	Explicit negative stereotypes about Indians (continuous scale, 0-1)
`impl.prejud`	Implicit bias (IAT score) against Indian Americans (continuous scale, 0-1)
`h1bvis.supp`	Support for increasing H-1B visas (5-point scale, 0-1)
`indimm.supp`	Support for increasing Indian immigration (5-point scale, 0-1)

The main outcome of interest (`h1bvis.supp`) – your dependent variable – was measured by the following survey item: "Some people have proposed that the U.S. government should increase the number of H-1B visas, which are allowances for U.S. companies to hire workers from foreign countries to work in highly skilled occupations (such as engineering, computer programming, and high-technology). Do you think the U.S. should increase, decrease, or keep about the same number of H-1B visas?" Another outcome (`indimm.supp`) specifically asked respondents about their feelings toward "the number of immigrants from India." Both variables have the following response options: `0` = "decrease a great deal", `0.25` = "decrease a little", `0.5` = "keep about the same", `0.75` = "increase a little", `1` = "increase a great deal". In other words, they are ordered, and we'll treat them as pseudo-continuous variables in our analysis (with the assumption that the underlying latent construct is continuous even though we only measure it on a 5-point response scale); alternatively, we could model them as ordered categorical outcomes, but that is beyond the scope of today's assignment).

To measure explicit stereotypes (`expl.prejud`), respondents were asked to evaluate Indians on a series of traits: capable, polite, hardworking, hygienic, and trustworthy. All responses were then used to create a scale from `0` (only positive traits of Indians) to `1` (no positive traits of Indians). Implicit bias (`impl.prejud`) is measured via the *Implicit Association Test* (IAT) which is designed to gauge the strength of associations linking social categories (e.g., European vs Indians) to evaluative anchors (e.g., good vs bad). Individual who are prejudiced against Indians should be quicker at making classifications of faces and words when *European American* (*Indian American*) is paired with *good* (*bad*) than when *European American* (*Indian American*) is paired with *bad* (*good*). If you want, you can test yourself here.

Question 1

After setting your working directory in R, import the data (making sure that it has loaded correctly) as an object called `immig`. (Hint: If you load `tidyverse`, then you can use `glimpse()` to view the data, variable names, and first few observations.)

```
## Load necessary packages
pacman::p_load(expss, ggplot2, tidyverse)

## Import the data (assuming it is saved in a folder called 'data')
immig <- read.csv("data/immig.csv")
head(immig)  # Check the first few rows of data
```

```
##   h1bvis.supp indimm.supp employed female age expl.prejud impl.prejud
## 1        1.00        1.00        1      1  31        0.30   0.3372782
## 2        0.75        0.75        0      0  64        0.35   0.5695007
## 3        0.25        0.25        0      1  69        0.25   0.5966338
## 4        0.00        0.00        0      0  49        0.85   0.7398279
## 5        0.00        0.25        0      1  51          NA          NA
## 6        0.50        0.25        1      1  61        0.25   0.4412750
##   nontech.whitcol tech.whitcol
## 1               1            0
## 2               0            0
## 3               0            0
## 4               0            0
## 5               0            0
## 6               0            0
```

```
glimpse(immig)  # Check the data the tidyverse way
```

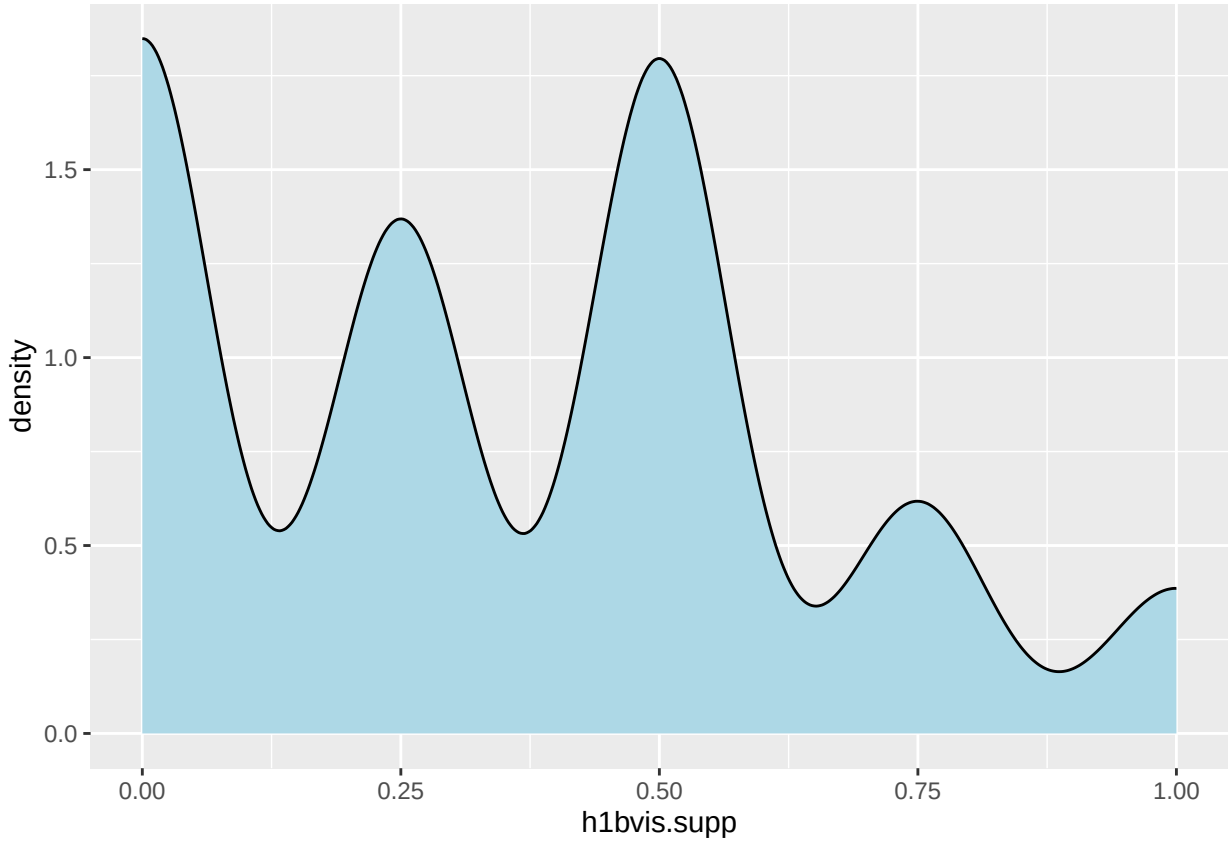
```
## Rows: 1,133
## Columns: 9
## $ h1bvis.supp     <dbl> 1.00, 0.75, 0.25, 0.00, 0.00, 0.50, 0.00, 0.50, 1.00, ~
## $ indimm.supp     <dbl> 1.00, 0.75, 0.25, 0.00, 0.25, 0.25, 0.00, 0.00, 0.00, ~
```

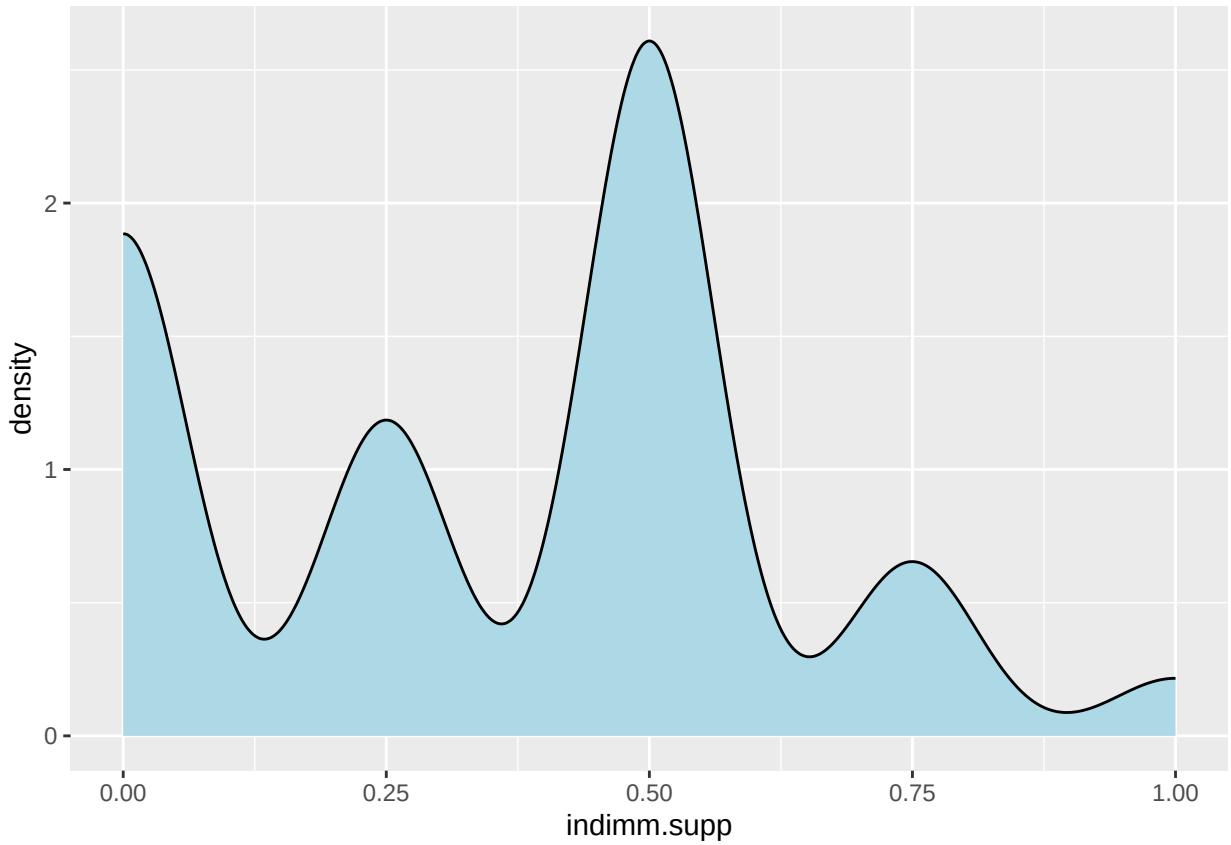
```
## $ employed        <int> 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 0, ~
## $ female          <int> 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 1, 0, 1, ~
## $ age             <int> 31, 64, 69, 49, 51, 61, 64, 47, 67, 43, 43, 50, 42, 39~
## $ expl.prejud     <dbl> 0.30, 0.35, 0.25, 0.85, NA, 0.25, NA, 0.45, 0.55, NA, ~
## $ impl.prejud     <dbl> 0.3372782, 0.5695007, 0.5966338, 0.7398279, NA, 0.4412~
## $ nontech.whitcol <int> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, ~
## $ tech.whitcol    <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
```

Now let's start by examining the frequency of responses to the immigration attitude questions (i.e., our outcomes, or dependent variables). What's the proportion of people who are willing to increase the quota for high-skilled foreign professionals (`h1bvis.supp`)? How about support for immigration from India (`indimm.supp`)? (Hint: you can get the proportions by adding the `prop.table()` function to a frequency table; e.g., `prop.table( table(df$variable) )`, or you can obtain those using the `fre()` function from the 'expss' package). It is also a good idea to visualize these outcomes using a density plot: `ggplot(immig, aes(h1bvis.supp) ) + geom_density(fill = 'lightblue)`. Don't forget to add labels! What do you see?

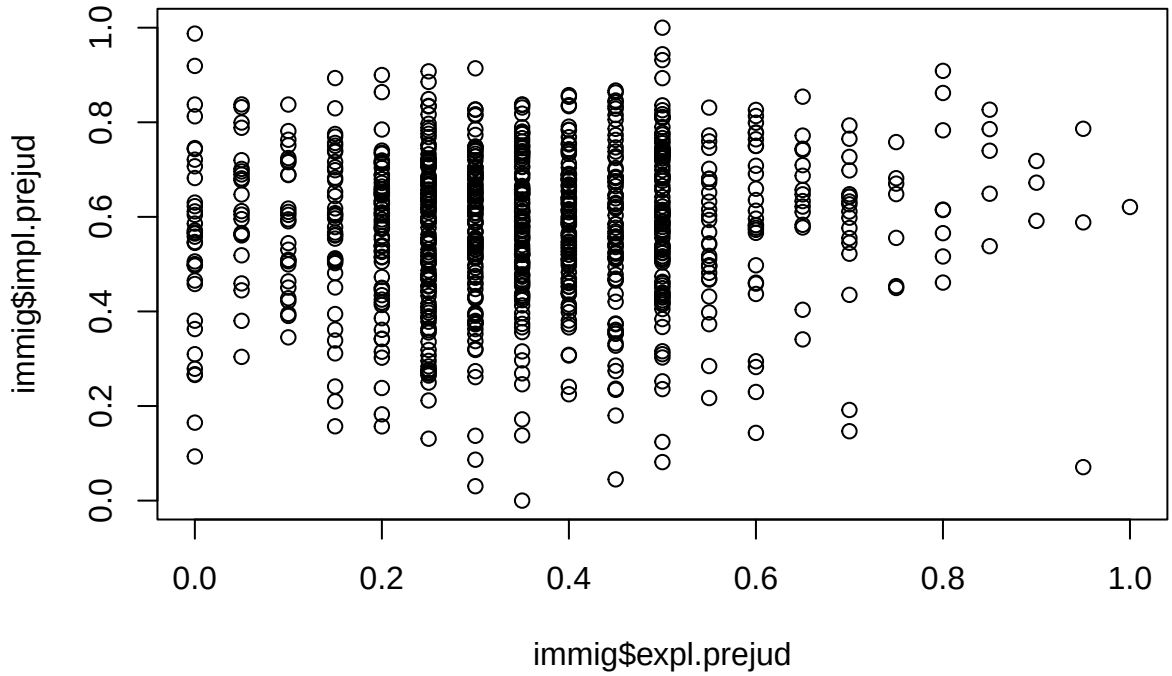
```
##
##          0       0.25        0.5       0.75          1
## 0.30748663 0.22727273 0.29857398 0.10249554 0.06417112
```

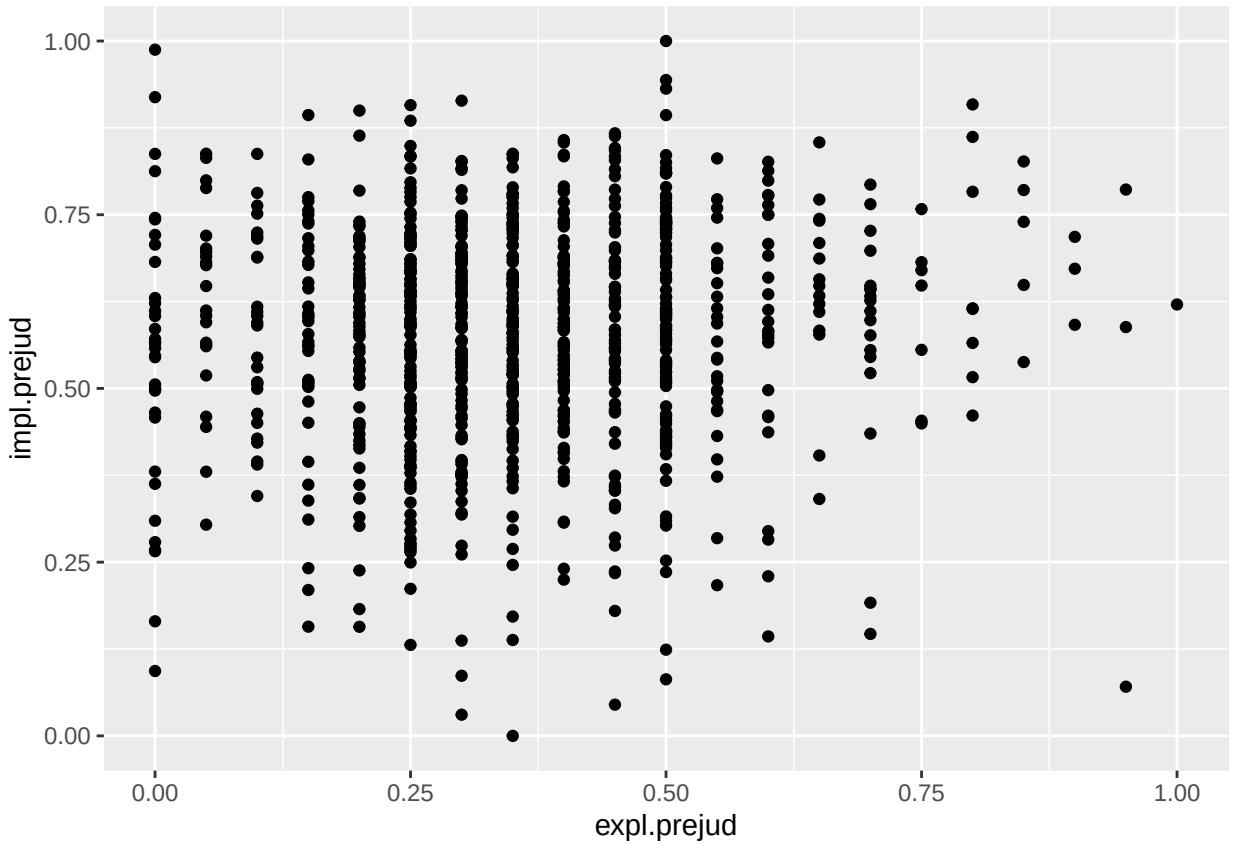
```
##
##          0       0.25        0.5       0.75          1
## 0.28787879 0.18092692 0.39839572 0.09982175 0.03297683
```



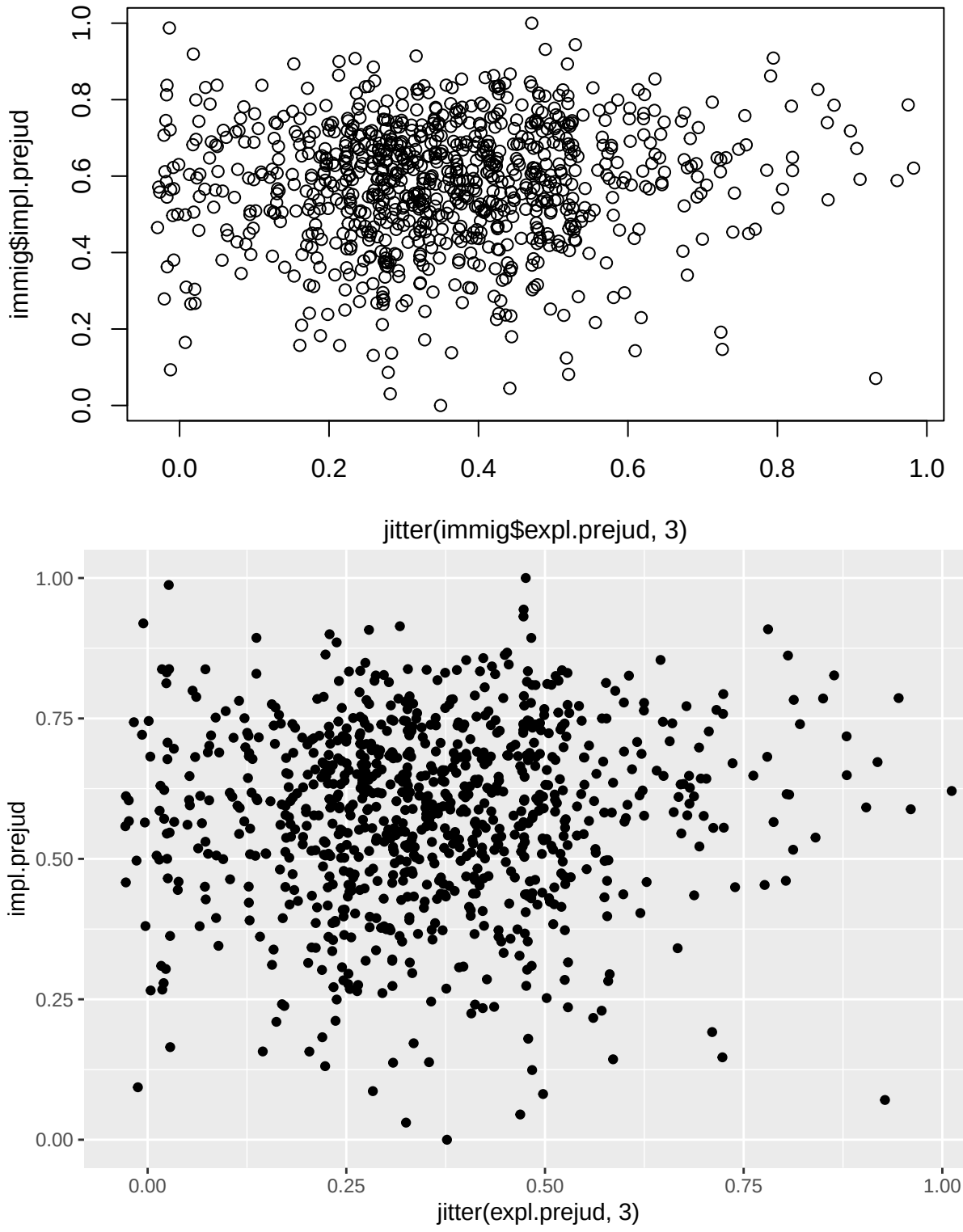


Now let's create a scatterplot of the two different measures of cultural threat. You can do this using the `plot()` function: `plot(immig$expl.prejud, immig$impl.prejud)`. Alternatively, you can create these using the `ggplot2` package: `ggplot(immig, aes(expl.prejud, impl.prejud) ) + geom_point()`. Put explicit stereotyping about Indians (`expl.prejud`) on the x-axis and implicit bias against Indian Americans (`impl.prejud`) on the y-axis.

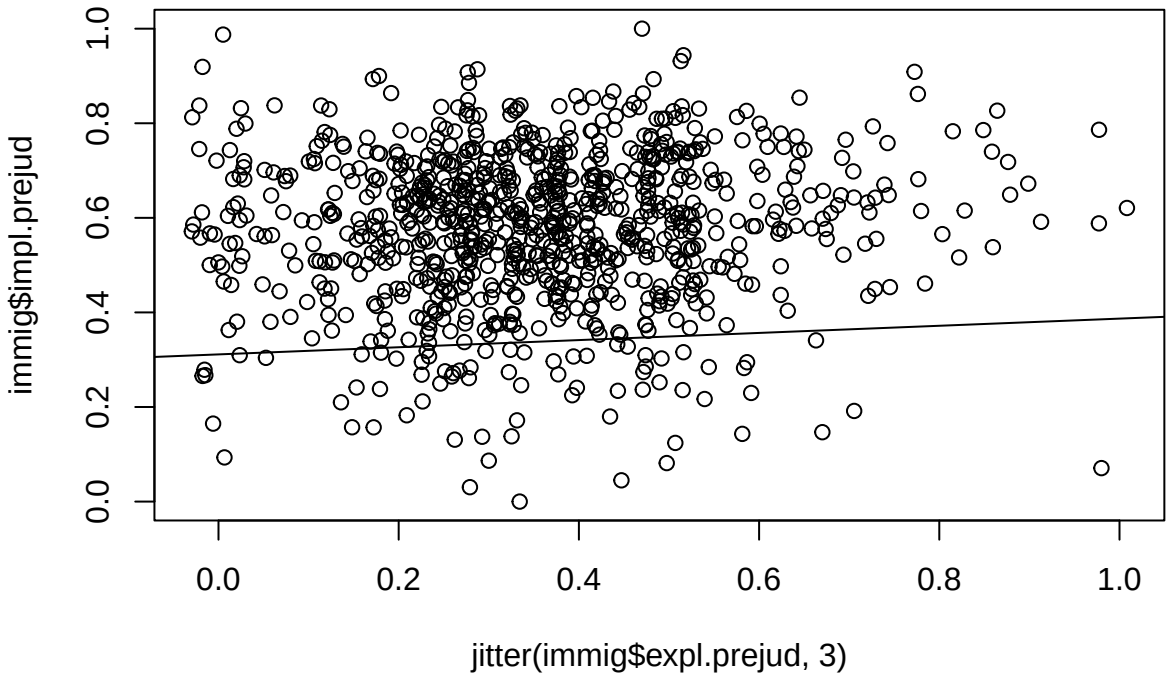




Notice that the scatterplot looks a bit odd because explicit prejudice isn't continuous. Once again, we can combine functions to improve the aesthetics of the plot. Let's try adding a small amount of random noise around the values of this variable to make it appear more continuous than it is. The `jitter()` function in R can help us do this; e.g., `plot( jitter(immig$expl.prejud, 3), immig$impl.prejud)` (the '3' tells R how much random noise to add; try different values beginning with `1` and see what happens!). How does the new scatterplot look? Again, don't forget the labels!

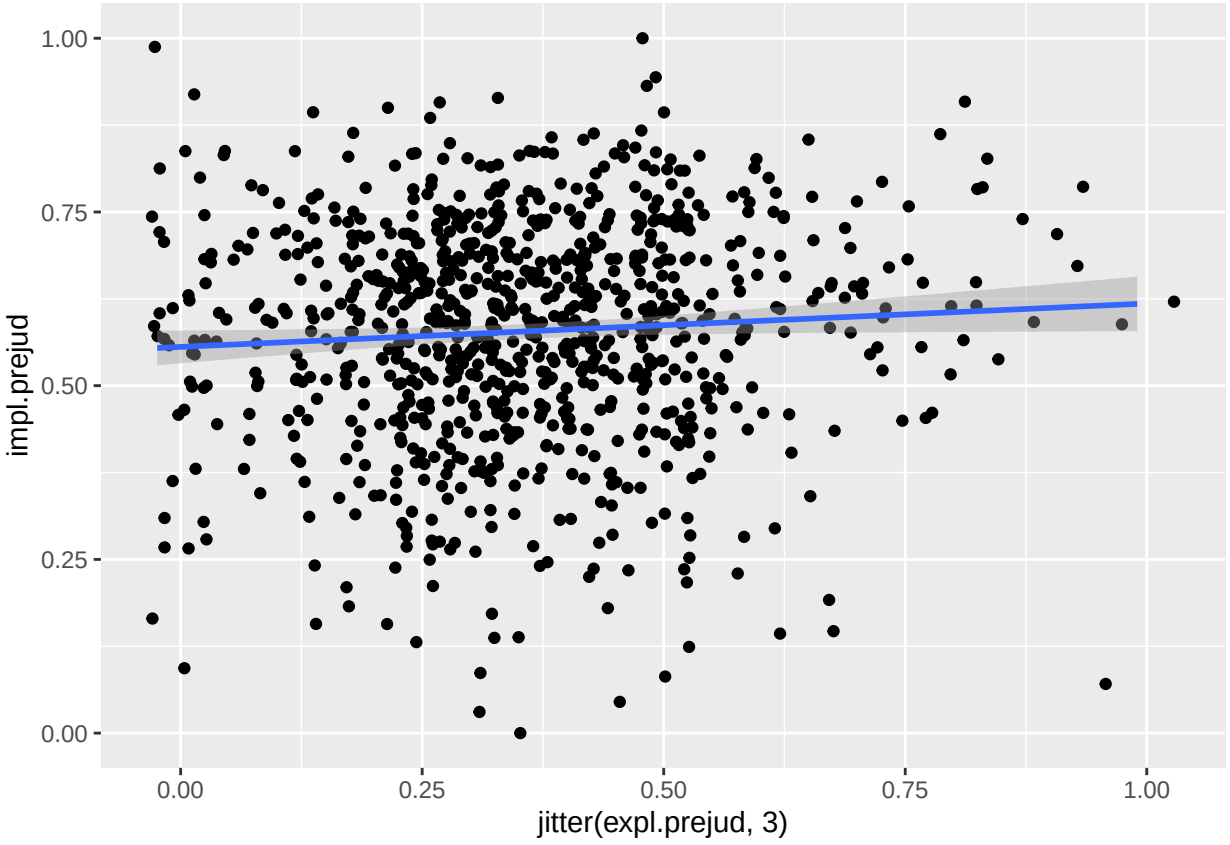


Finally, let's add a linear regression line to the scatterplot (the curly braces are included to ensure the plot works in R Markdown). You can do this by using the `lm()` function to conduct the regression: `model1 <- lm(expl.prejud ~ impl.prejud, data = immig)`. Add the results from your stored model using the `abline()` function after you've created your plot using the `plot()` function; e.g., `abline(model1)`, assuming your regression results are stored as `model1`. As always, don't forget to label your plot.



If you've created your scatterplot using `ggplot()`, then the fitted regression line can be added in a slightly different manner: `ggplot(immig, aes(expl.prejud, impl.prejud) ) + geom_point() + geom_smooth(method = lm) # Add the fitted regression line`

```
## `geom_smooth()` using formula = 'y ~ x'
```

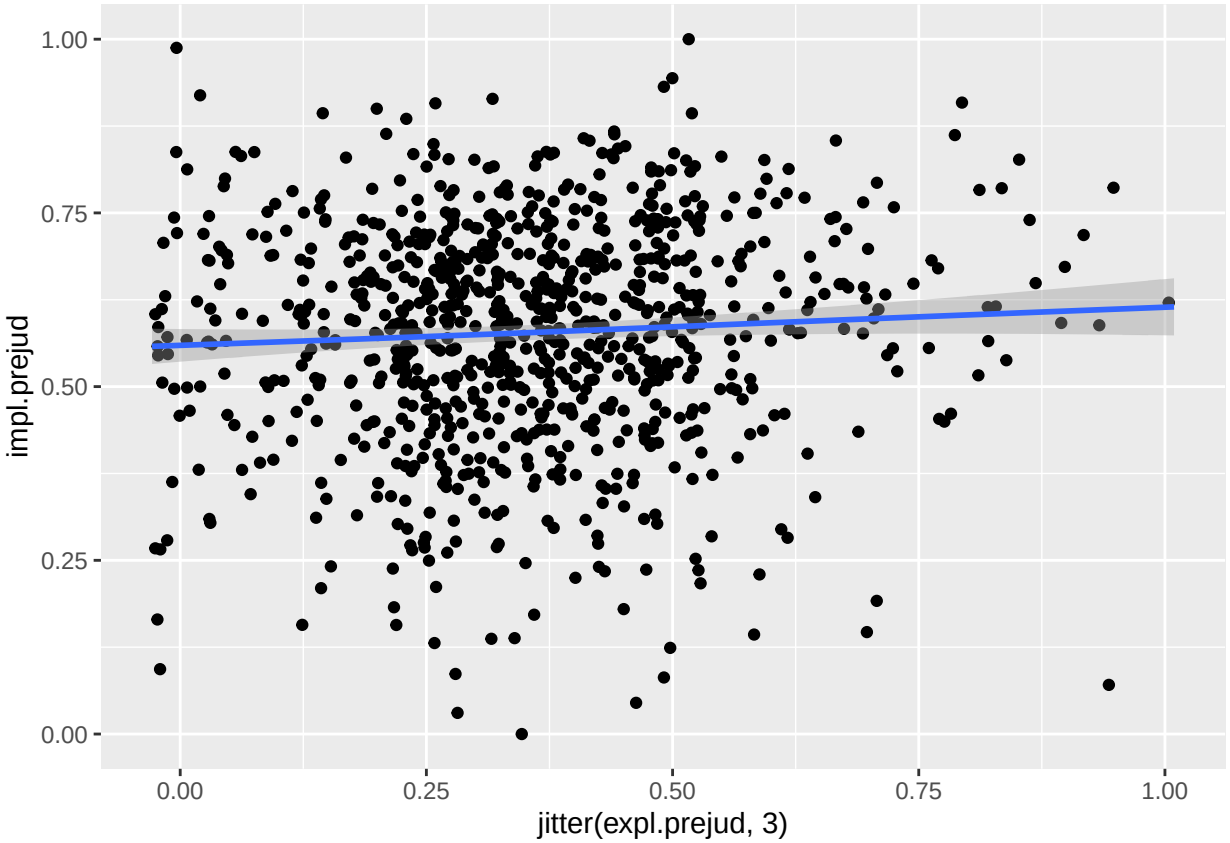


As an aside, non-linear trends can be visualized using `ggplot` as we've done above, but by removing the `method = lm` argument. These are called 'Loess curves' and uses local weighted regression to fit a smoothed curve through the data points in a scatterplot:

```
ggplot(immig, aes(expl.prejud, impl.prejud) ) +   geom_point() +   geom_smooth() # Add the Loess curve
```

However, in this case, the Loess curve shows a fairly linear fit to the data.

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



If you're feeling adventurous, calculate the correlation between these variables using the `cor()` function: `cor(immig$expl.prejud, immig$impl.prejud, use = "complete.obs")`. Based on these results, what can you say about the relationship between explicit and implicit prejudice? Is this surprising to you?

```
## [1] 0.06864096
```

Question 2

Now let's test the labor market argument using linear regression. If the economic threat hypothesis is correct, opposition to H-1B visas should be more pronounced among those who are economically threatened by this policy such as individuals in the high-technology sector. Let's regress `h1bvis.supp` on the `tech.whitcol` dummy variable to test whether this is the case: `model2a <- lm(h1bvis.supp ~ tech.whitcol, data = immig)`. Print the results using the `summary()` function: `summary(model2a)`. Interpret the results. What does a 1-unit change in `tech.whitcol` (i.e., moving from non-tech jobs to high-tech jobs) mean for your outcome (i.e., what is the coefficient for `tech.whitcol`)?

```
##
## Call:
## lm(formula = h1bvis.supp ~ tech.whitcol, data = immig)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.34995 -0.34995 -0.09995  0.15005  0.70339
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.349953   0.009211  37.994   <2e-16 ***
## tech.whitcol -0.053343   0.040166  -1.328    0.184
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3003 on 1120 degrees of freedom
##   (11 observations deleted due to missingness)
## Multiple R-squared:  0.001572,   Adjusted R-squared:  0.0006808
## F-statistic: 1.764 on 1 and 1120 DF,  p-value: 0.1844
```

At the same time, tech workers shouldn't really be worried about *general* Indian immigration because as a group, they don't pose a specific *economic* threat. Regress `indimm.supp` on the `tech.whitcol` dummy variable and interpret the results as you've done above. Store these regression results in an object called `model2b`.

```
##
## Call:
## lm(formula = indimm.supp ~ tech.whitcol, data = immig)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.3525 -0.3525  0.1475  0.1475  0.6525
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.352540   0.008460  41.674   <2e-16 ***
## tech.whitcol -0.005082   0.036891  -0.138     0.89
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2758 on 1120 degrees of freedom
##   (11 observations deleted due to missingness)
## Multiple R-squared:  1.695e-05,  Adjusted R-squared:  -0.0008759
## F-statistic: 0.01898 on 1 and 1120 DF,  p-value: 0.8904
```

Do the results from these two regression models support the economic threat hypothesis? Why or why not?

Question 3

When testing hypotheses, it's important to have an appropriate comparison group. We could argue that comparing tech workers to everybody else as we did in Question 2 may be problematic due to a variety of confounding factors (such as respondents' skill level and employment status). This means that our regression model may be misspecified, and thus, our estimates will be biased.

Let's create a new *factor* variable called `group` with the following assignments:

Value	Description
Tech WC‘	employed in the technology industry
Non-Tech WC‘	employed in non-tech “white-collar” jobs

Value	Description
Other‘	employed in any other sector
Unemployed‘	unemployed

Factors are categorical variables that can be ordered or unordered. For instance, the immigration outcomes are technically ordered categories (though they've been given numerical values in the dataset):

Numerical Value	Descriptive Category Label
0‘	decrease a great deal
0.25‘	decrease a little
0.5‘	keep about the same
0.75‘	increase a little
1‘	increase a great deal

Thus, we could've used these response options without numerical values if we had told R that it was an *ordered factor* (and specified what the ordering was). For example, we could use the `factor()` function (this is for demonstration purposes only, no need to try to run this code!):

In R's memory, this factor is represented by the numbers 1, 2, 3, 4, and 5, though we see the actual category labels, which makes it easier to work with for humans (who needs a codebook now?!).

Another added benefit of factor variables is that R will *automagically* dummy code them for use in a regression model (rather than having to create *m - 1* dummy variables for *m* categories). A dummy variable is just a variable that takes values of 1 or 0 depending on whether that category attribute is present or absent. Moreover, we can change the reference category by changing the `levels` argument (R will assume the first category should be the reference category, which may not be want we want for interpretation; each coefficient for a dummy variable tells us the increase or decrease in y associated with a 1-unit increase in X, or in this case, moving from the reference category to the value indicated by that category for which an estimate is provided).

There are different ways to accomplish this task, below is how you might do this using base R (which is ugly and clunky):

```
immig$group <- NA
immig$group[immig$tech.whitcol == 1 & immig$nontech.whitcol == 0
            & immig$employed == 1] <- "Tech WC"

immig$group[immig$tech.whitcol == 0 & immig$nontech.whitcol == 1
            & immig$employed == 1] <- "Non-Tech WC"

immig$group[immig$tech.whitcol == 0 & immig$nontech.whitcol == 0
            & immig$employed == 1] <- "Other"

immig$group[immig$employed == 0] <- "Unemployed"

immig$group <- factor(immig$group,
                          levels = c("Tech WC", "Non-Tech WC", "Other", "Unemployed"),
                          ordered = FALSE)
levels(immig$group)
```

But we can make this code much easier to read using `tidyverse`, which is why people recommend using it. Here, we're going to use `mutate()` to add a new variable to the dataframe, and the `case_when()` function to use conditional logic. Typically, we might use `ifelse()` (or `if_else()` using `tidyverse` convetions) work

really well when a single set of conditions must be satisfied. However, if we need to use multiple or complex conditions, then it requires us to nest if-else statements, which become cumbersome to read and are prone to errors. The `case_when()` function allows you to separate logic line-by-line into readable code:

```
immig <-  # Store the operations back into the 'immig' object
        immig %>%   # Pass this object to functions using %>%
        mutate(group = case_when(  # Use conditional logic (if this ~ assign to that)
                tech.whitcol == 1 & nontech.whitcol == 0 & employed == 1 ~ "Tech WC",
                tech.whitcol == 0 & nontech.whitcol == 1 & employed == 1 ~ "Non-Tech WC",
                tech.whitcol == 0 & nontech.whitcol == 0 & employed == 1 ~ "Other",

                tech.whitcol == 0 & nontech.whitcol == 0 & employed == 0 ~ "Unemployed"
                ),
               group = factor(group, levels = c("Tech WC", "Non-Tech WC", "Other", "Unemployed")
                              )
        )

is.factor(immig$group)  # Confirm that the new variable is a factor
```

```
## [1] TRUE
```

Using your newly created `group` unordered factor variable, compare support for H-1B visas across working conditions using a linear regression approach (tech workers should serve as the reference category). That is: `model3 <- lm(h1bvis.supp ~ group, data = immig)`.

```
##
## Call:
## lm(formula = h1bvis.supp ~ group, data = immig)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.39224 -0.34607 -0.09607  0.15393  0.70339
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       0.29661    0.03911   7.584 7.02e-14 ***
## groupNon-Tech WC  0.09563    0.05555   1.722   0.0854 .
## groupOther        0.04946    0.04149   1.192   0.2334
## groupUnemployed   0.05217    0.04121   1.266   0.2058
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3004 on 1118 degrees of freedom
##   (11 observations deleted due to missingness)
## Multiple R-squared:  0.002675,   Adjusted R-squared:  -1.234e-06
## F-statistic: 0.9995 on 3 and 1118 DF,  p-value: 0.3923
```

Interpret the results. Does the pattern of results from Question 2 hold? Is there evidence for the economic threat hypothesis now that you've looked at the results from this regression model?

Question 4

Now, we could argue that people who work in the tech sector are disproportionately young and male, which may further confound our results. To account for this possibility, fit another linear regression model with `h1bvis.supp` as the outcome and `group` as the predictor, but this time include other *control* variables such as `age` and `gender` (labelled `female` because it is dummy coded so that female respondents = 1, male

respondents = 0). Does adding these additional predictors to the model change the results and, if so, how? Interpret each of the estimates from the model.

```
##
## Call:
## lm(formula = indimm.supp ~ group + female + age, data = immig)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47682 -0.29488  0.07579  0.17608  0.69414
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        0.470230   0.044228  10.632  < 2e-16 ***
## groupNon-Tech WC   0.070031   0.050896   1.376    0.169
## groupOther         0.017648   0.038091   0.463    0.643
## groupUnemployed    0.048305   0.037968   1.272    0.204
## female            -0.072248   0.017247  -4.189 3.02e-05 ***
## age               -0.002195   0.000554  -3.963 7.86e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2723 on 1116 degrees of freedom
##   (11 observations deleted due to missingness)
## Multiple R-squared:  0.02844,    Adjusted R-squared:  0.02409
## F-statistic: 6.534 on 5 and 1116 DF,  p-value: 5.318e-06
```

Question 5

Let's fit a linear regression model with `h1bvis.supp` (as your outcome) regressed on all threat indicators (`group`, `expl.prejud`, `impl.prejud`), and obtain its R^2 value (again using the `summary()` function for the saved model results). How much variation in the outcome is explained by the model (R^2 ranges from 0 to 1, and it will provide you with the proportion of variance explained by your model)? Based on the model fit, what can you conclude about the role of threat factors?

```
##
## Call:
## lm(formula = h1bvis.supp ~ group + impl.prejud + expl.prejud,
##     data = immig)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4961 -0.2740 -0.0393  0.1762  0.7603
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        0.50239    0.05789   8.678  < 2e-16 ***
## groupNon-Tech WC   0.10270    0.06191   1.659  0.09749 .
## groupOther         0.04719    0.04567   1.033  0.30181
## groupUnemployed    0.04099    0.04537   0.904  0.36645
## impl.prejud       -0.19169    0.06136  -3.124  0.00184 **
## expl.prejud       -0.25268    0.05555  -4.548 6.16e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.2951 on 889 degrees of freedom
##   (238 observations deleted due to missingness)
## Multiple R-squared:  0.03834,    Adjusted R-squared:  0.03293
## F-statistic: 7.088 on 5 and 889 DF,  p-value: 1.641e-06
```

Question 6

Besides economic and cultural threat, many scholars also argue that gender is an important predictor of immigration attitudes. While there is some evidence that women are slightly less opposed to immigration than men, it may also be true that gender moderates the effect of factors like cultural threat (e.g., women feel less culturally threatened by immigrants than men). To determine whether this is the case, fit a linear regression of H-1B support on the *interaction* between gender (`female`) and implicit prejudice (`impl.prejud`). An interaction will test whether the values of one variable (e.g., `female`) vary at levels of another variable (e.g., `impl.prejud`). In other words, is the slope coefficient the same for men and women? If not, then the coefficient for the interaction will be non-zero, and the hypothesis test of this estimate will reach conventional levels of statistical significance – the *probability value*, or *p-value* for short, will be less than .05.

The *p-value* tells you the probability of finding a value as extreme as (or greater than) what we observed in the data (if there were actually no relationship, or a zero value, in the population). The lower the p-value, the less likely the result is due to chance alone.

One criticism of p-values is that in large samples, many estimates from the analysis will be statistically significant simply because the denominator, or bottom number, is increasingly large. Researchers often use p-values as a shortcut for determining whether there is sufficient evidence to support a particular conclusion. A p-value smaller than .05 tells us that there's less than a .05 probability (or 5% chance) of that result popping up randomly. There's nothing inherently special about $p < .05$ – it's just an arbitrary rule of thumb to help us determine whether the result will generalise to the population. We could just as easily specify that $p < .01$ or $p < .001$, and so on, but there's a tradeoff between the error of finding something that is a result of chance and failing to find something that is actually there. If our p-value threshold is too high, we're likely to do the former; too low and we're likely to do the latter. Confused? So are many people who learn about p-values. But the important thing to remember is that a p-value isn't the final answer but helps us decide whether a finding is likely to exist in the real world.

Of course, whether a result is statistically significant tells us nothing about its effect size. In contrast, practical significance refers to whether a result has a meaningful impact on the outcome of interest. For instance, a study may include a statistically significant finding at the $p < .05$ level, yet the effect size associated with the finding may be quite small, sometimes even approaching zero.

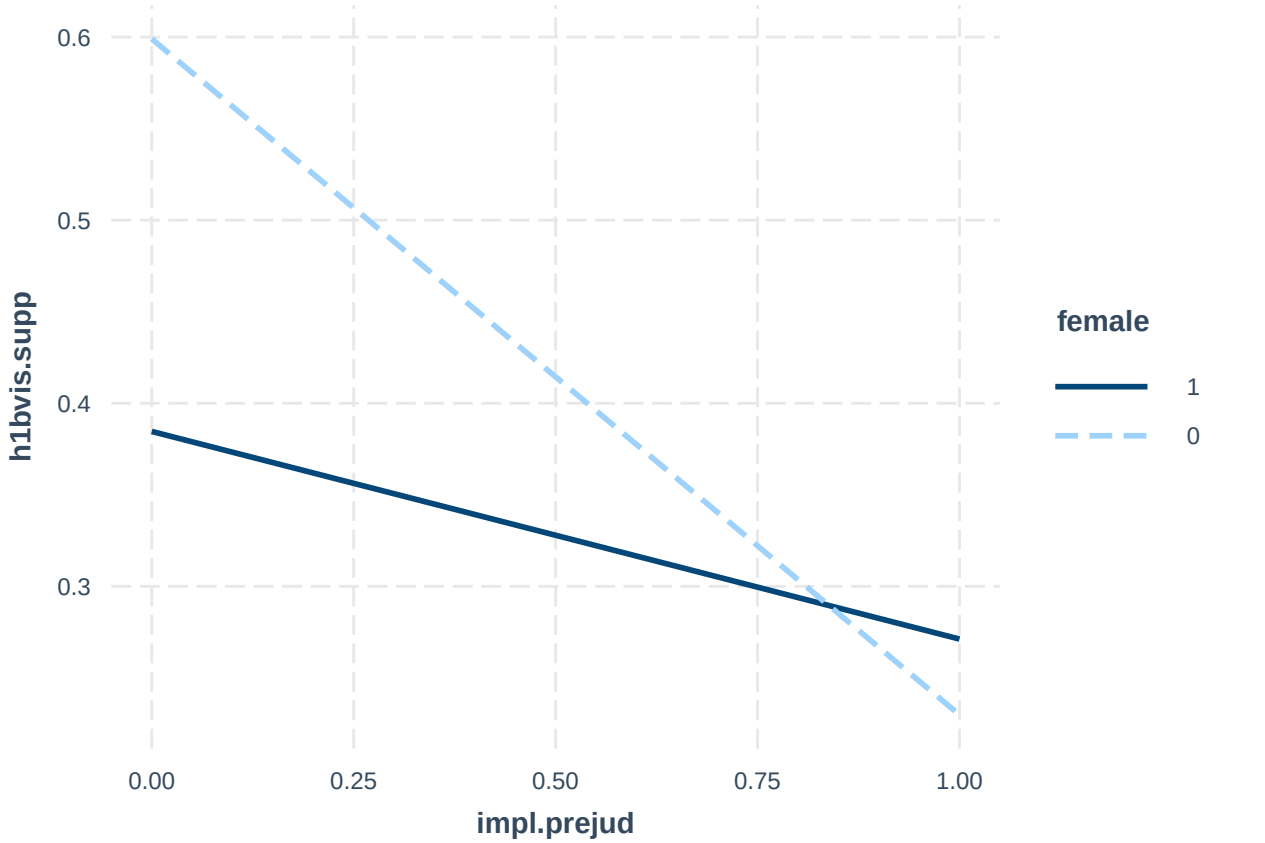
Anyway, you can model an interaction by multiplying the predictors together like this `female*impl.prejud` so your full regression model would be: `model6 <- lm(h1bvis.supp ~ female*impl.prejud, data = immig)`, which is equivalent to writing it out completely: `model6 <- lm(h1bvis.supp ~ female + impl.prejud + female*impl.prejud, data = immig)`. If the coefficient for the interaction statistically significant ($p < .05$), which tells us whether the slopes differ from one another – that is, the fitted regression lines are *not* parallel. Is the interaction statistically significant?

```
##
## Call:
## lm(formula = h1bvis.supp ~ female * impl.prejud, data = immig)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.58244 -0.30385 -0.05303  0.18450  0.72116
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         0.59903    0.05476  10.939  < 2e-16 ***
## female             -0.21443    0.06995  -3.066  0.00223 **
```

```
## impl.prejud        -0.36923    0.09312  -3.965 7.86e-05 ***
## female:impl.prejud  0.25582    0.11788   2.170  0.03024 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2954 on 1005 degrees of freedom
##   (124 observations deleted due to missingness)
## Multiple R-squared:  0.03086,    Adjusted R-squared:  0.02797
## F-statistic: 10.67 on 3 and 1005 DF,  p-value: 6.615e-07
```

Interactions are difficult to understand without plotting them visually. Plot the predicted level of H-1B support (on the y-axis) across the range of implicit bias (on the x-axis) by levels of gender. You can do this quite easily with a package called `interactions` using the `interact_plot()` function:

```
pacman::p_load(interactions)
model6.plot <- interact_plot(model6, pred = impl.prejud, modx = female)
model6.plot  # Display the plot
```



Considering the results from your plot, would you agree that gender *moderates* the relationship between cultural threat and immigration attitudes?

If you can get through this lab exercise, then you can complete what most social scientists who work with quantitative data do for a living. Bravo (because this is no mean feat)!